File Processing

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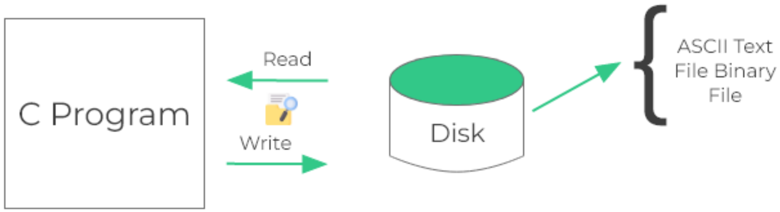
File Introduction

- Files are places where data can be stored permanently.
- Some programs expect the same set of data to be fed as input every time it is run.
  - Cumbersome.
  - Better if the data are kept in a file, and the program reads from the file.

Text File vs Binary File

File Handling in PASCAL

Based on "ASCII code"



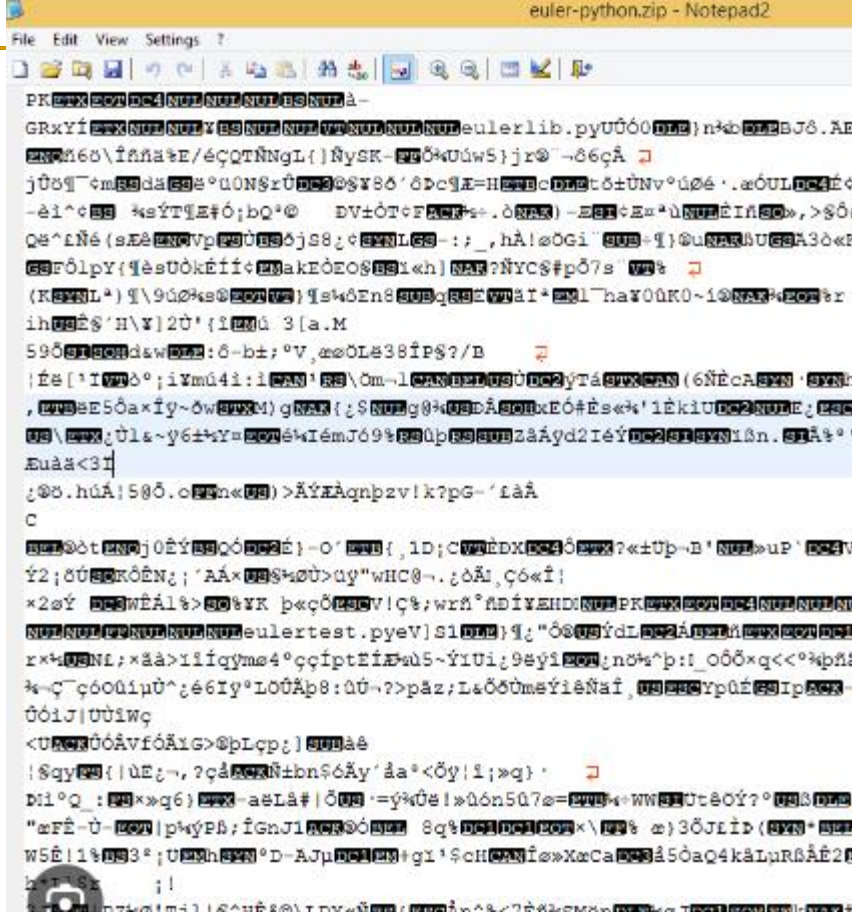
Text Files	Binary Files
• When using a text file, we write out separately each of the pieces of data about a given record. • The text file will be readable by an editor	• When using a binary file we write whole record data to the file at once. • but the numbers in the binary file will not be readable in this way.

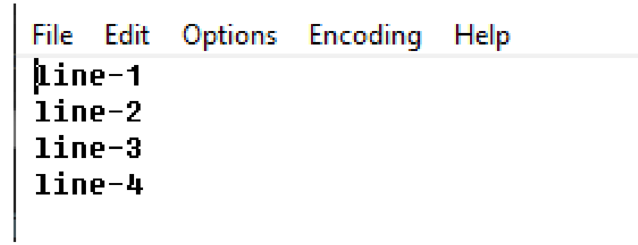
Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

Source: www.LookupTables.com

ASCII Code	Char
128	Ç
129	ü
130	é
131	â
132	ä
133	à
134	å
135	ç
136	ê
137	ë
138	è
139	ï
140	î
141	ì
142	Ä
143	Å
144	É
145	§
146	Æ
147	ô
148	ö
149	ò
150	û
151	ù
152	ÿ
153	Ö
154	Ü
155	¢
156	£
157	¥
158	₧
159	ƒ
160	á
161	í
162	ó
163	ú
164	ñ
165	Ñ
166	ª
167	º
168	¿
169	⌐
170	¬
171	½
172	¼
173	¡
174	«
175	»
176	░
177	▒
178	▓
179	│
180	┤
181	╡
182	╢
183	╖
184	╕
185	╣
186	║
187	╗
188	╝
189	╜
190	╛
191	┐
192	└
193	┴
194	┬
195	├
196	─
197	┼
198	╞
199	╟
200	╚
201	╔
202	╩
203	╦
204	╠
205	═
206	╬
207	╧
208	╨
209	╤
210	╥
211	╙
212	╘
213	╒
214	╓
215	╫
216	╪
217	┘
218	┌
219	█
220	▄
221	▌
222	▐
223	▀
224	α
225	ß
226	Γ
227	π
228	Σ
229	σ
230	µ
231	τ
232	Φ
233	Θ
234	Ω
235	δ
236	∞
237	φ
238	ε
239	∩
240	≡
241	±
242	≥
243	≤
244	⌠
245	⌡
246	÷
247	≈
248	°
249	∙
250	·
251	√
252	^{n}
253	^{2}
254	■
255	

te c chnology
Uni





Text File

readable

Binary File

unreadable

File Edit Options Encoding Help

```
line-1
line-2
line-3
line-4

--- SAMPLE 1
#include <iostream>

int main()
{
char ch;
FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS21
if (fp1 == NULL) {
printf("cannot open file !!");
return 0;
}
do {
long pos = ftell(fp1);
ch = getc(fp1);
printf("%c \t  %x \t %d \n",ch, ch , pos);
} while (ch != EOF);

fclose(fp1);

}
```

TEXT FILE

ปกติจะมีรหัส 0xD, 0xA
เพื่อเป็นตัวควบคุมการเว้นบรรทัด

File Edit Options Encoding Help

```
00000000: 6C 69 6E 65 2D 31 0D 0A|6C 69 6E 65 2D 32 0D 0A  | line-1..line-2..
00000010: 6C 69 6E 65 2D 33 0D 0A|6C 69 6E 65 2D 34 0D 0A  | line-3..line-4..
00000020: 0D 0A 2D 2D 2D 20 53 41|4D 50 4C 45 20 31 0D 0A  | ..--- SAMPLE 1..
00000030: 23 69 6E 63 6C 75 64 65|20 3C 69 6F 73 74 72 65  | #include <iostre
00000040: 61 6D 3E 0D 0A 0D 0A 69|6E 74 20 6D 61 69 6E 28  | am>....int main(
00000050: 29 0D 0A 7B 0D 0A 63 68|61 72 20 63 68 3B 0D 0A  | )..{..char ch;..
00000060: 46 49 4C 45 20 2A 66 70|31 20 3D 20 66 6F 70 65  | FILE *fp1 = fope
00000070: 6E 28 20 22 44 3A 2F 6D|79 64 6F 63 2F 63 6F 73  | n( "D:/mydoc/cos
00000080: 32 31 30 31 2F 44 65 6D|6F 43 4F 53 32 31 30 31  | 2101/DemoCOS2101
00000090: 2F 44 65 62 75 67 2F 74|65 73 74 31 2E 74 78 74  | /Debug/test1.txt
000000A0: 22 2C 20 22 72 74 22 29|3B 0D 0A 69 66 20 28 66  | ", "rt");..if (f
000000B0: 70 31 20 3D 3D 20 4E 55|4C 4C 29 20 7B 0D 0A 70  | p1 == NULL) {..p
000000C0: 72 69 6E 74 66 28 22 63|61 6E 6E 6F 74 20 6F 70  | rintf("cannot op
000000D0: 65 6E 20 66 69 6C 65 20|21 21 22 29 3B 0D 0A 72  | en file !!");..r
000000E0: 65 74 75 72 6E 20 30 3B|0D 0A 7D 0D 0A 64 6F 20  | eturn 0;..}..do 
000000F0: 7B 0D 0A 6C 6F 6E 67 20|70 6F 73 20 3D 20 66 74  | {..long pos = ft
00000100: 65 6C 6C 28 66 70 31 29|3B 0D 0A 63 68 20 3D 20  | ell(fp1);..ch = 
00000110: 67 65 74 63 28 66 70 31|29 3B 0D 0A 70 72 69 6E  | getc(fp1);..prin
00000120: 74 66 28 22 25 63 20 5C|74 20 20 25 78 20 5C 74  | tf("%c \t  %x \t
00000130: 20 25 64 20 5C 6E 22 2C|63 68 2C 20 63 68 20 2C  |  %d \n",ch, ch ,
00000140: 20 70 6F 73 29 3B 0D 0A|7D 20 77 68 69 6C 65 20  |  pos);..} while 
00000150: 28 63 68 20 21 3D 20 45|4F 46 29 3B 0D 0A 0D 0A  | (ch != EOF);....
00000160: 66 63 6C 6F 73 65 28 66|70 31 29 3B 0D 0A 0D 0A  | fclose(fp1);....
00000170: 7D 0D 0A 0D 0A 2D 2D 20|53 41 4D 50 4C 45 20 32  | }....-- SAMPLE 2
00000180: 0D 0A 23 64 65 66 69 6E|65 20 5F 43 52 54 5F 53  | ..#define _CRT_S
00000190: 45 43 55 52 45 5F 4E 4F|5F 57 41 52 4E 49 4E 47  | ECURE_NO_WARNING
000001A0: 53 0D 0A 23 69 6E 63 6C|75 64 65 20 3C 69 6F 73  | S..#include <ios
000001B0: 74 72 65 61 6D 3E 0D 0A|0D 0A 0D 0A 69 6E 74 20  | tream>......int 
000001C0: 6D 61 69 6E 28 29 0D 0A|7B 0D 0A 63 68 61 72 20  | main()..{..char 
000001D0: 63 68 3B 0D 0A 46 49 4C|45 20 2A 66 70 31 20 3D  | ch;..FILE *fp1 =
000001E0: 20 66 6F 70 65 6E 28 22|44 3A 2F 6D 79 64 6F 63  |  fopen("D:/mydoc
000001F0: 2F 63 6F 73 32 31 30 31|2F 44 65 6D 6F 43 4F 53  | /cos2101/DemoCOS
00000200: 32 31 30 31 2F 44 65 62|75 67 2F 74 65 73 74 31  | 2101/Debug/test1
00000210: 2E 74 78 74 22 2C 20 22|72 74 22 29 3B 0D 0A 69  | .txt", "rt");..i
00000220: 66 20 28 66 70 31 20 3D|3D 20 4E 55 4C 4C 29 20  | f (fp1 == NULL) 
00000230: 7B 0D 0A 70 72 69 6E 74|66 28 22 63 61 6E 6E 6F  | {..printf("canno
00000240: 74 20 6F 70 65 6E 20 66|69 6C 65 20 21 21 22 29  | t open file !!")
00000250: 3B 0D 0A 72 65 74 75 72|6E 20 30 3B 0D 0A 7D 0D  | ;..return 0;..}.
00000260: 0A 0D 0A 64 6F 20 7B 0D|0A 6C 6F 6E 67 20 70 6F  | ...do {..lonq po
```

File Edit Options Encoding Help

```
%PDF-1.5
%ḿḿḿḿ
1 0 obj
<</Type/Catalog/Pages 2 0 R/Lang(en-US) /StructTre
true>>/Metadata 1478 0 R/ViewerPreferences 1479 0 
endobj
2 0 obj
<</Type/Pages/Count 11/Kids[ 3 0 R 20 0 R 38 0 R 4
R 58 0 R 60 0 R 62 0 R] >>
endobj
3 0 obj
<</Type/Page/Parent 2 0 R/Resources<</ExtGState<</
R>>/Font<</F1 6 0 R/F2 12 0 R/F3 14 0 R/F4 16 0 R/
R>>/ProcSet[/PDF/Text/ImageB/ImageC/ImageI] >>/Med
0 R/Group<</Type/Group/S/Transparency/CS/DeviceRGB
endobj
4 0 obj
<</Filter/FlateDecode/Length 889>>
stream
[illegible]
endstream
endobj
5 0 obj
<</Type/ExtGState/BM/Normal/ca 1>>
endobj
6 0 obj
<</Type/Font/Subtype/Type0/BaseFont/BCDEEE+THSarabunPSK-Bold/Encoding/Identity-H/
DescendantFonts 7 0 R/ToUnicode 1454 0 R>>
endobj
```

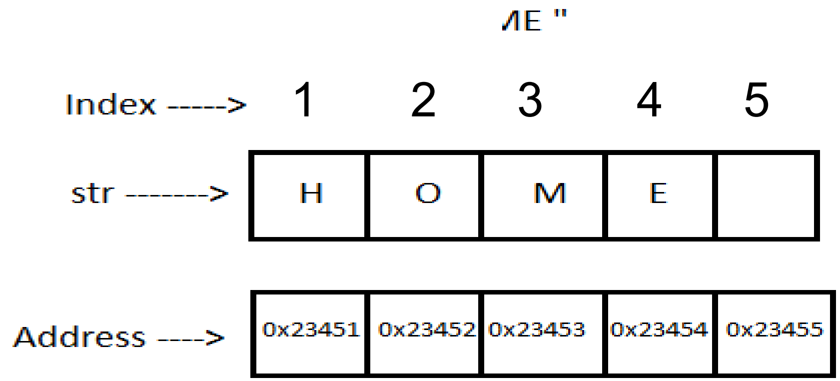
File Edit Options Encoding Help

```
00000000: 25 50 44 46 2D 31 2E 35|0D 0A 25 B5 B5 B5 B5 0D | %PDF-1.5..%ḿḿḿḿ.
00000010: 0A 31 20 30 20 6F 62 6A|0D 0A 3C 3C 2F 54 79 70 | .1 0 obj..<</Typ
00000020: 65 2F 43 61 74 61 6C 6F|67 2F 50 61 67 65 73 20 | e/Catalog/Pages 
00000030: 32 20 30 20 52 2F 4C 61|6E 67 28 65 6E 2D 55 53 | 2 0 R/Lang(en-US
00000040: 29 20 2F 53 74 72 75 63|74 54 72 65 65 52 6F 6F | ) /StructTreeRoo
00000050: 74 20 36 34 20 30 20 52|2F 4D 61 72 6B 49 6E 66 | t 64 0 R/MarkInf
00000060: 6F 3C 3C 2F 4D 61 72 6B|65 64 20 74 72 75 65 3E | o<</Marked true>
00000070: 3E 2F 4D 65 74 61 64 61|74 61 20 31 34 37 38 20 | >/Metadata 1478 
00000080: 30 20 52 2F 56 69 65 77|65 72 50 72 65 66 65 72 | 0 R/ViewerPrefer
00000090: 65 6E 63 65 73 20 31 34|37 39 20 30 20 52 3E 3E | ences 1479 0 R>>
000000A0: 0D 0A 65 6E 64 6F 62 6A|0D 0A 32 20 30 20 6F 62 | ..endobj..2 0 ob
000000B0: 6A 0D 0A 3C 3C 2F 54 79|70 65 2F 50 61 67 65 73 | j..<</Type/Pages
000000C0: 2F 43 6F 75 6E 74 20 31|31 2F 4B 69 64 73 5B 20 | /Count 11/Kids[ 
000000D0: 33 20 30 20 52 20 32 30|20 30 20 52 20 33 38 20 | 3 0 R 20 0 R 38 
000000E0: 30 20 52 20 34 30 20 30|20 52 20 34 34 20 30 20 | 0 R 40 0 R 44 0 
000000F0: 52 20 35 31 20 30 20 52|20 35 34 20 30 20 52 20 | R 51 0 R 54 0 R 
00000100: 35 36 20 30 20 52 20 35|38 20 30 20 52 20 36 30 | 56 0 R 58 0 R 60
00000110: 20 30 20 52 20 36 32 20|30 20 52 5D 20 3E 3E 0D |  0 R 62 0 R] >>.
00000120: 0A 65 6E 64 6F 62 6A 0D|0A 33 20 30 20 6F 62 6A | .endobj..3 0 obj
00000130: 0D 0A 3C 3C 2F 54 79 70|65 2F 50 61 67 65 2F 50 | ..<</Type/Page/P
00000140: 61 72 65 6E 74 20 32 20|30 20 52 2F 52 65 73 6F | arent 2 0 R/Reso
00000150: 75 72 63 65 73 3C 3C 2F|45 78 74 47 53 74 61 74 | urces<</ExtGStat
00000160: 65 3C 3C 2F 47 53 35 20|35 20 30 20 52 2F 47 53 | e<</GS5 5 0 R/GS
00000170: 31 31 20 31 31 20 30 20|52 3E 3E 2F 46 6F 6E 74 | 11 11 0 R>>/Font
00000180: 3C 3C 2F 46 31 20 36 20|30 20 52 2F 46 32 20 31 | <</F1 6 0 R/F2 1
00000190: 32 20 30 20 52 2F 46 33|20 31 34 20 30 20 52 2F | 2 0 R/F3 14 0 R/
000001A0: 46 34 20 31 36 20 30 20|52 2F 46 35 20 31 38 20 | F4 16 0 R/F5 18 
000001B0: 30 20 52 3E 3E 2F 50 72|6F 63 53 65 74 5B 2F 50 | 0 R>>/ProcSet[/P
000001C0: 44 46 2F 54 65 78 74 2F|49 6D 61 67 65 42 2F 49 | DF/Text/ImageB/I
000001D0: 6D 61 67 65 43 2F 49 6D|61 67 65 49 5D 20 3E 3E | mageC/ImageI] >>
000001E0: 2F 4D 65 64 69 61 42 6F|78 5B 20 30 20 30 20 36 | /MediaBox[ 0 0 6
000001F0: 31 32 20 37 39 32 5D 20|2F 43 6F 6E 74 65 6E 74 | 12 792] /Content
00000200: 73 20 34 20 30 20 52 2F|47 72 6F 75 70 3C 3C 2F | s 4 0 R/Group<</
00000210: 54 79 70 65 2F 47 72 6F|75 70 2F 53 2F 54 72 61 | Type/Group/S/Tra
00000220: 6E 73 70 61 72 65 6E 63|79 2F 43 53 2F 44 65 76 | nsparency/CS/Dev
00000230: 69 63 65 52 47 42 3E 3E|2F 54 61 62 73 2F 53 2F | iceRGB>>/Tabs/S/
00000240: 53 74 72 75 63 74 50 61|72 65 6E 74 73 20 30 3E | StructParents 0>
00000250: 3E 0D 0A 65 6E 64 6F 62|6A 0D 0A 34 20 30 20 6F | >..endobj..4 0 o
```

การอ่านข้อมูลจาก array of character

- Array of character

Type
    Index: Integer
    c1,c2,c3:character
    str : string

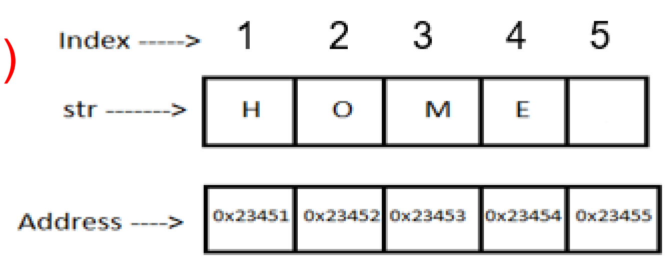


str := “HOME”;
Index := 1;

index identifies location of elements on array to do such as read and write, called access.

c1 := str[index]; — Read ‘H’
index := index+1; — adjust position forward (2)

c2 = str[index]; — Read ‘O’
index := index+1; — adjust position forward (3)



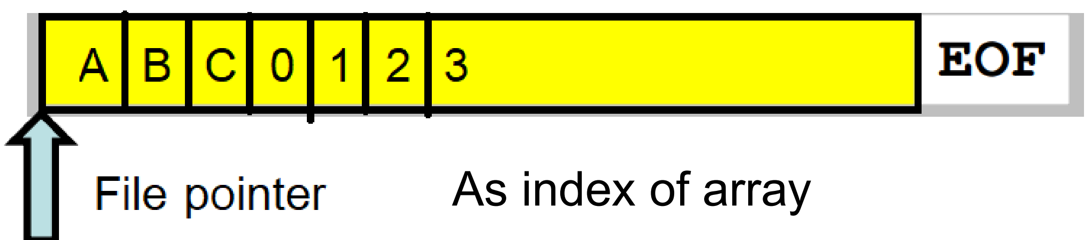
Index := 4;

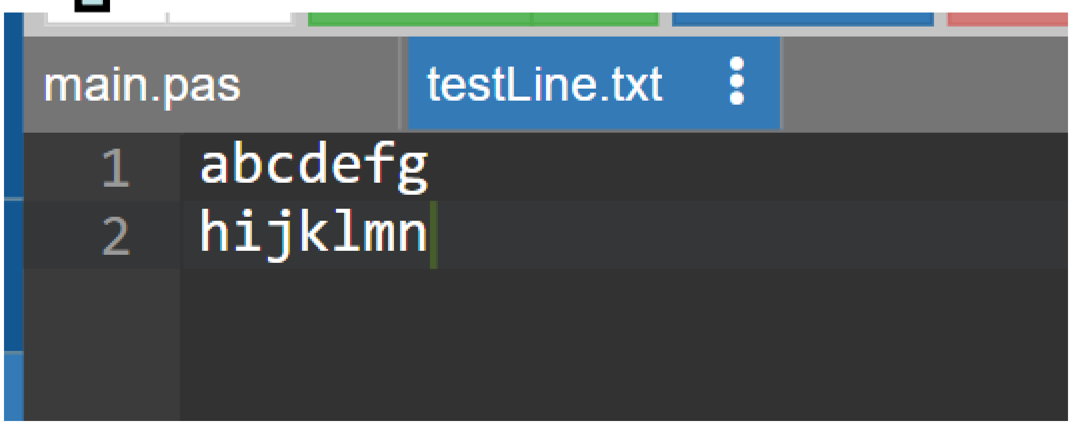
c3 = str[index] — Write ‘E’
index := index+1; — adjust position backward

File Concepts

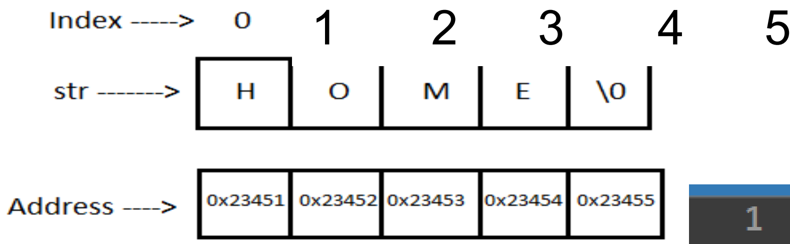
1. มอง file เป็น array of character
2. มี File pointer /Find Handle ทำหน้าที่เป็นตัวชี้ตำแหน่งข้อมูลปัจจุบัน
3. ทุกครั้งที่มีการอ่าน(read/write) file pointer จะขยับมา 1 ตัวอักษร

File could be viewed as array of character





testLine.txt



readtestLine.pas

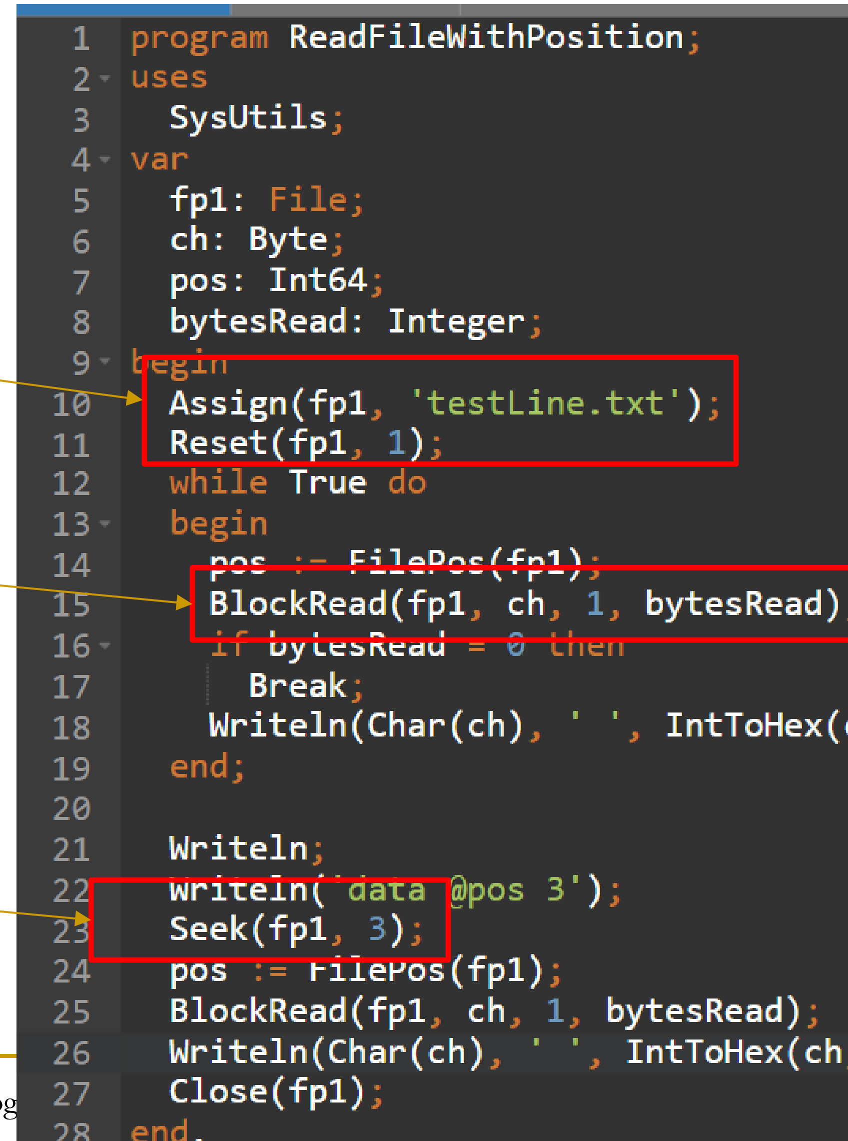
Type
Index: Integer
c1,c2,c3:character
str : string

str := “HOME”;
Index := 1;

c1 := str[index]; Read ‘H’
index := index+1; adjust position forward (2)

c2 = str[index]; Read ‘O’
index := index+1; adjust position forward (3)

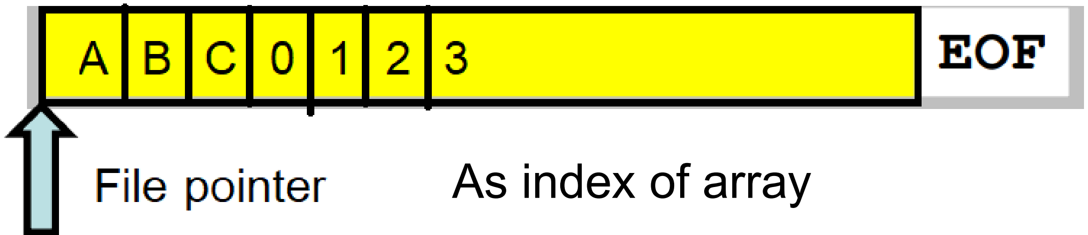
Index := 4;
c3 = str[index] read ‘E’
index := index+1; adjust position backward



```
program ReadFileWithPosition;
uses
  SysUtils;
var
  fp1: File;
  ch: Byte;
  pos: Int64;
  bytesRead: Integer;
begin
  Assign(fp1, 'testLine.txt');
  Reset(fp1, 1);
  while True do
  begin
    pos := FilePos(fp1);
    BlockRead(fp1, ch, 1, bytesRead)
    if bytesRead = 0 then
      Break;
    Writeln(Char(ch), ' ', IntToHex(
  end;

  Writeln;
  Writeln('data @pos 3');
  Seek(fp1, 3);
  pos := FilePos(fp1);
  BlockRead(fp1, ch, 1, bytesRead);
  Writeln(Char(ch), ' ', IntToHex(ch
  Close(fp1);
end.
```

File could be viewed as array of character



readtestLine.pas

```pascal
program ReadFileWithPosition;
uses
  SysUtils;
var
  fp1: File;
  ch: Byte;
  pos: Int64;
  bytesRead: Integer;
begin
  Assign(fp1, 'testLine.txt');
  Reset(fp1, 1);
  while True do
  begin
    pos := FilePos(fp1);
    BlockRead(fp1, ch, 1, bytesRead);
    if bytesRead = 0 then
      Break;
    Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
  end;

  Writeln;
  Writeln('data @pos 3');
  Seek(fp1, 3);
  pos := FilePos(fp1);
  BlockRead(fp1, ch, 1, bytesRead);
  Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
  Close(fp1);
end.
```

```
Target OS: Linux fo
Compiling main.pas
Linking a.out
33 lines compiled,
0 30 0
1 31 1
2 32 2
3 33 3
4 34 4
5 35 5
6 36 6
7 37 7
8 38 8
9 39 9
0 30 10
a 61 11
b 62 12
c 63 13
d 64 14
e 65 15
f 66 16
g 67 17

data @pos 3
3 33 3
```

Array vs File concepts

- Array of character

Type
    Index: Integer
    c1,c2:character
    str : string

str := “HOME”;
Index := 1;

c1 := str[index];
index := index+1;

c2 = str[index];
index := index+1;

read

adjust position forward

index := 3
str[index] = ‘A’
index := index+1;

write

adjust position backward

```
program ReadFileWithPosition;
uses
  SysUtils;
var
  fp1: File;
  ch: Byte;
  pos: Int64;
  bytesRead: Integer;
begin
  Assign(fp1, 'testLine.txt');
  Reset(fp1, 1);
  while True do
  begin
    pos := FilePos(fp1);
    BlockRead(fp1, ch, 1, bytesRe
    if bytesRead = 0 then
      Break;
    Writeln(Char(ch), ' ', IntToH
  end;

  Writeln;
  Writeln('data @pos 3');
  Seek(fp1, 3);
  pos := FilePos(fp1);
  BlockRead(fp1, ch, 1, bytesRead
  Writeln(Char(ch), ' ', IntToHex
  Close(fp1);
end.
```

ลำดับการทำงานแฟ้มข้อมูล

1. เปิดแฟ้มข้อมูล	**1.1. Assign (f, 'myfile.dat')**
	Reset (f) /Rewrite (f) /Append(f)
2. การจัดการบนแฟ้มข้อมูล	
2.1. อ่าน	**2.1. BlockRead() ,Read()**
2.2. เขียน	**2.2.** BlockWrite(), **Write(), Append()**
2.3. การสืบค้นตำแหน่ง	**2.3. Seek() , FilePost ( )**
3. ปิดแฟ้มข้อมูล	**3. close( )**

Text File Manage

คำสั่งการจัดการแฟ้มแบ่งออกเป็น 3 กลุ่ม (Based on FPC)

- Text File Mode (TEXT)
- Binary File Mode (FILE)

ลำดับ operation บน text file

0: สร้าง file pointer	0. var f : Text
1. กำหนดชื่อแฟ้ม	1. assign(f,’mydata’)
2. เปิดแฟ้มข้อมูล	2.1. Reset(f) / เปิดเพื่ออ่าน
	2.1. Append(f) / เปิดเขียนแบบต่อท้าย
	2.2. Rewrite(f) / เปิดเขียนแบบ overwrite
3. การจัดการบนแฟ้มข้อมูล	
3.1. อ่าน	3.1 Read(f, var1, var2, ...);
3.2. เขียน	ReadLn(f, var1, var2, ...);
~~3.3. การสืบค้นตำแหน่ง~~	3.2 WriteLn(f, 'New line’);
~~(ทำไม่ได้)~~	Write(f, ‘line’);
4. ปิดแฟ้มข้อมูล	4. Close(f);

Opening/Close a File

- A file must be "opened" before it can be used.

```
Var   fi : Text

Assign( fi, 'ReadFile');
Reset( fi ) ;



Close( fi ) ;
```

```
Var   fi : Text

Assign( fi, 'overwriteFile');
Rewrite( fi ) ;



Close(fi) ;
```

```
Var   fi : Text

Assign( fi, 'AppendFile');
Append( fi ) ;



Close( fi ) ;
```

Files and Streams

- Read/Readln : ต้องมีรูปแบบ

```
Var
        fi:text
        str3: string[3+1]
        i1      : integer;
        f       : real;
        str2  : string[2+1];
        i2      : integer

    Readln(  fi , str1 ,   i ,  f      , str2 , i2 );
                     A01 123  40.5  B1   13
                     A02 124  40.1  B2   14
```

Files and Streams

- Read/Readln :

  1. ต้องมีรูปแบบข้อมูลเหมือนกันทั้ง file

     A01 123 40.5 B1 13

     A02 124 40.1 B2 14

  2. แต่ละ field มีขนาดคงที่ มี space ขั้น

  3. กรณี field มีชนิดเป็น string

     3.1. ขนาดของ field ต้องเท่ากันทุกๆ record

     3.2. ต้องจองพื้นที่ของตัวแปรมีขนาดเท่ากับ field+1

  เช่น A02 -> ตัวแปรที่อ่านต้องมีขนาด string[3+1];

  B1 -> ตัวแปรที่อ่านต้องมีขนาด string[2+1];

main.pas | teacherInfo.txt

```pascal
program ReadTeacherInfo;

var
  f: Text;
  id: string[9+1]; //add 1 for spc
  no: Integer;
  level: string[1+1]; //add 1 for spc
  grade: real;
  score: Integer;
  code: string[4+1]; //add 1 for spc
  group:Integer;
//T65001001 1 A 1.1 10 A001 -3
begin
  Assign(f, 'teacherInfo.txt');
  Reset(f);   { เปิดไฟล์เพื่ออ่าน }

  while not Eof(f) do
  begin
    ReadLn(f, id, no,level, grade,score,code,group);

    Writeln('ID     : ', id);
    Writeln('No     : ', no);
    Writeln('level    : ', level);
    Writeln('Grade : ', grade:0:2);
    Writeln('Score : ', score);
    Writeln('Code  : ', code);
    Writeln('group  : ', group);
    Writeln('----------------------');
  end;

  Close(f);   { ปิดไฟล์ }
end.
```

reafmt.pas

```
T65001001 1 A 1.1 10 A001 -3
T65001002 2 B 2.1 20 A002 2
T65001003 3 C 3.1 20 A003 -1
T65001004 4 D 4.1 40 A004 0
T65001005 5 A 5.1 50 A005 -1
T65001006 6 A 6.1 60 A006 3
T65001007 7 A  7.1 70 A007 2
T65001008 8 B 8.1 80 A008 1
```

teacherInfo.txt

Write Operations on Files

Write/WriteLn

var fi:Text;

Write(fi, var1,‘ ’ , “const-string” , ‘ ’ , 1, ‘ ’ , 3.2);

main.pas | teacherInfo.txt | teacherInfo.out

writfmt.pas

```pascal
program ReadTeacherInfo;

var
  f1,f2: Text;
  id: string[9+1]; //add 1 for spc
  no: Integer;
  level: string[1+1]; //add 1 for spc
  grade: real;
  score: Integer;
  code: string[4+1]; //add 1 for spc
  group:Integer;
//T65001001 1 A 1.1 10 A001 -3
begin
  Assign(f1, 'teacherInfo.txt');
  Reset(f1);   { เปิดไฟล์เพื่ออ่าน }

  Assign(f2, 'teacherInfo.out');
  //Rewrite(f2);
  Append(f2);

  while not Eof(f1) do
  begin
    ReadLn(f1, id, no,level, grade,score,code,group);

    Writeln('ID    : ', id);
    Writeln('No    : ', no);
    Writeln('level    : ', level);
    Writeln('Grade : ', grade:0:2);
    Writeln('Score : ', score);
    Writeln('Code  : ', code);
    Writeln('group  : ', group);
    Writeln('----------------------');
    if group > 0 then
      Writeln(f2, id,' ', no,' ',level,' ', grade:0:2,' ',score,' ',code,' ',group);
  end;

  Close(f1);   { ปิดไฟล์ }
  Close(f2);
end.
```

Operation on File

- Read Only
- Create a new File or over write exist File
- Read/Write File (update file)
- Append File

main.pas | teacherInfo.txt | class1.txt | class2.txt | class3.out

```pascal
program MergeTwoFiles;
var
  fileA, fileB, fileC: Text;
  num1, num2: Integer;
  f1, f2: Boolean;

begin
  Assign(fileA, 'class1.txt');
  Reset(fileA);

  Assign(fileB, 'class2.txt');
  Reset(fileB);

  Assign(fileC, 'class3.out');
  Rewrite(fileC);

  Read(fileA, num1);
  Read(fileB, num2);

  while not Eof(fileA) and not Eof(fileB) do
  begin
    if num1 < num2 then
    begin
      Writeln(fileC, num1);
      Read(fileA, num1);
    end
    else if num2 < num1 then
    begin
      Writeln(fileC, num2);
        Read(fileB, num2);
    end
    else
    begin
      Writeln(fileC, num1);
      Read(fileA, num1);
      Read(fileB, num2);
    end;
  end;

  Close(fileA);
  Close(fileB);
  Close(fileC);
end.
```

Lister - [I

File Edit

```
1
3
5
6
7
9
11
13
12
```

File Edi

```
2
4
6
8
10
12
14
16
```

simmerg.pas

จากปัญหา

ตัวอย่าง 1

สมมุติว่าข้อมูล อาจารย์คณะวิทยาศาสตร์มี 150 ดังนี้

เพศ (sex) : ชาย = 1 , หญิง = 2

อายุ (age) : ? ปี

สถานภาพสมรส (status) : โสด = 1 , แต่งงาน = 2 , อื่นๆ = 2

ปริญญาสูง (degree) : ตรี = 1 , โท = 2 , เอก = 3

จำนวนปีที่สอน (experience) : ? ปี

จงเขียน

1. โปรแกรมรับข้อมูลของอาจารย์คณะวิทยาศาสตร์จำนวน 150 คน
2. หาจำนวนปีเฉลี่ยของ อายุ และ จำนวนปีที่สอน

AVEREAGE AGE = xx.xx Years

AVEREAGE Experience = XX.XX Years

StruEmpAvg.pas

main.pas

```pascal
program EmployeeAverage;
const numPerson = 3; //ค่าคงที่
type
  Employee = record
    sex: Integer;
    age: Integer;
    stat: Integer;
    degr: Integer;
    tyear: Integer;
  end;

var
  person: array[1..numPerson] of Employee;
  i: Integer;
  avg_age, avg_expr: Real;
  numAgeAbove, numYearAbove: Integer;

begin
  { Input data for first 2 employees }
  for i := 1 to numPerson do
  begin
    Write('sex(1-2) = ');   ReadLn(person[i].sex);
    Write('age = ');   ReadLn(person[i].age);
    Write('stat (1-3) = ');  ReadLn(person[i].stat);
    Write('degr(1-3) = ');  ReadLn(person[i].degr);
    Write('tyear = '); ReadLn(person[i].tyear);
  end;

  avg_age := 0;
  avg_expr := 0;

  { Compute sums }
  for i := 1 to numPerson do
  begin
    avg_age := avg_age + person[i].age;
    avg_expr := avg_expr + person[i].tyear;
  end;

  avg_age := avg_age / numPerson;
  avg_expr := avg_expr / numPerson;

  numAgeAbove := 0;
  numYearAbove := 0;

  { Count how many are above average }
  for i := 1 to numPerson do
  begin
    if person[i].age > avg_age then
      numAgeAbove := numAgeAbove + 1;

    if person[i].tyear > avg_expr then
      numYearAbove := numYearAbove + 1;
  end;

  { Output results }
  WriteLn('AVG AGE  = ', avg_age:0:2);
  WriteLn('NUMBER OF AGE ABOVE AVERAGE = ', numAgeAbove);
  WriteLn('AVG EXPR = ', avg_expr:0:2);
  WriteLn('NUMBER OF YEAR ABOVE AVERAGE = ', numYearAbove);
end.
```

ตัวอย่าง 1

สมมุติว่าข้อมูล อาจารย์คณะวิทยาศาสตร์มี 150 ดังนี้

เพศ (sex) : ชาย = 1 , หญิง = 2

อายุ (age) : ? ปี

สถานภาพสมรส (status) : โสด = 1 , แต่งงาน = 2 , อื่นๆ = 2

ปริญญาสูง (degree) : ตรี = 1 , โท = 2 , เอก = 3

จำนวนปีที่สอน (experience) : ? ปี

จงเขียน

1. โปรแกรมรับข้อมูลของอาจารย์คณะวิทยาศาสตร์จำนวน 150 คน
2. หาจำนวนปีเฉลี่ยของ อายุ และ จำนวนปีที่สอน

AVEREAGE AGE = xx.xx Years

AVEREAGE Experience = XX.XX Years

T65001001, รหัสอาจารย์ (2)
1, เพศ
20, จำนวนปีที่สอน
1, ระดับปริญญา
1 สถานภาพสมรศ

ให้ นักศึกษา เขียนโปรแกรมอ่านข้อมูลอาจารย์จากแฟ้ม (file) ซึ่งมีรูปแบบ (format) การจัดเก็บดังนี้

T65001001 1 20 1 1
T65001002 2 22 2 2
T65001003 1 21 2 2
T65001004 2 22 1 2
T65001005 2 23 3 2
T65001006 2 19 3 2

empData.txt

main.pas | empData.txt | class1.txt | class2.txt | class3.out | teacherInfo.txt

FileEmpAvg.pas

```
program TeacherInfo;
type
  employee = record
    Tcode : string[9+1];
    sex   : Integer;
    tyear : Integer;
    degr  : Integer;
    stat  : Integer;
  end;

var
  person : array[1..150] of employee;
  f        : Text;
  i, count : Integer;
  avg_expr : Real;

begin
  Assign(f, 'empData.txt');
  Reset(f);

  Writeln('code':12, 'sex':6, 'year':6, 'degr':6, 'stat':6);

  count := 0;
```

empData.txt

FileEmpAvg.pas

```
  while not Eof(f) do
  begin

    count := count+1;

    ReadLn(f,
           person[count].Tcode,
           person[count].sex,
           person[count].tyear,
           person[count].degr,
           person[count].stat);

    Writeln(person[count].Tcode:12,
            person[count].sex:6,
            person[count].tyear:6,
            person[count].degr:6,
            person[count].stat:6);
  end;

  Close(f);

  { คำนวณค่าเฉลี่ยปีที่สอน }
  avg_expr := 0;
  for i := 1 to count do
    avg_expr := avg_expr + person[i].tyear;

  if count > 0 then
    avg_expr := avg_expr / count;

  Writeln;
  Writeln('AVG EXPR = ', avg_expr:0:2);

end.
```

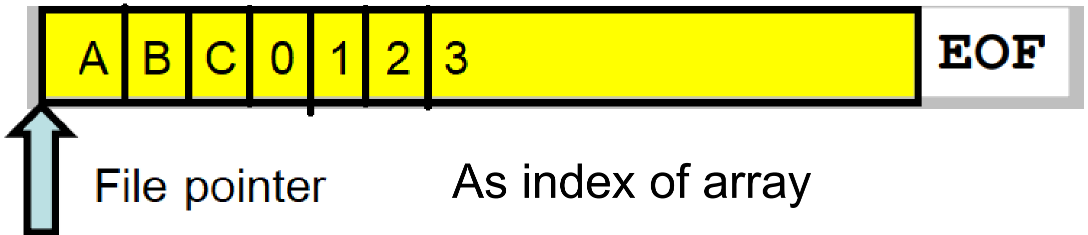
END

Binary File

ลำดับ operation บน text file

0: สร้าง file pointer	0. var f : Text
1. กำหนดชื่อแฟ้ม	1. assign(f,'mydata')
2. เปิดแฟ้มข้อมูล	2.1. Reset(f) / เปิดเพื่ออ่าน
	2.1. Append(f) / เปิดเขียนแบบต่อท้าย
	2.2. Rewrite(f) / เปิดเขียนแบบ overwite
3. การจัดการบนแฟ้มข้อมูล	
3.1. อ่าน	3.1 BlockRead (f, var1, ...);
3.2. เขียน	3.2 BlockWrite(f, var1,…);
3.3. การสืบค้นตำแหน่ง	FilePost(f) , Seek()
4. ปิดแฟ้มข้อมูล	4. Close(f);

File could be viewed as array of character



readtestLine.pas

```pascal
program ReadFileWithPosition;
uses
  SysUtils;
var
  fp1: File;
  ch: Byte;
  pos: Int64;
  bytesRead: Integer;
begin
  Assign(fp1, 'testLine.txt');
  Reset(fp1, 1);
  while True do
  begin
    pos := FilePos(fp1);
    BlockRead(fp1, ch, 1, bytesRead);
    if bytesRead = 0 then
      Break;
    Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
  end;

  Writeln;
  Writeln('data @pos 3');
  Seek(fp1, 3);
  pos := FilePos(fp1);
  BlockRead(fp1, ch, 1, bytesRead);
  Writeln(Char(ch), ' ', IntToHex(ch,2), ' ', pos);
  Close(fp1);
end.
```

```
Target OS: Linux fo
Compiling main.pas
Linking a.out
33 lines compiled,
0 30 0
1 31 1
2 32 2
3 33 3
4 34 4
5 35 5
6 36 6
7 37 7
8 38 8
9 39 9
0 30 10
a 61 11
b 62 12
c 63 13
d 64 14
e 65 15
f 66 16
g 67 17

data @pos 3
3 33 3
```

D:\mydoc\cos2101\DemoCOS2101\DemoCOS2101\Debug\simplereadfile.ex

rb

l	6c	0
i	69	1
n	6e	2
e	65	3
-	2d	4
1	31	5
	d	6
	a	7
l	6c	8
i	69	9
n	6e	10
e	65	11
-	2d	12
2	32	13
	d	14
	a	15
l	6c	16
i	69	17

rt

l	6c	0
i	69	1
n	6e	2
e	65	3
-	2d	4
1	31	5
	a	6
l	6c	8
i	69	9
n	6e	10
e	65	11
-	2d	12
2	32	13
	a	14
l	6c	16
i	69	17
n	6e	18
e	65	19

File Edit Options Encoding Help

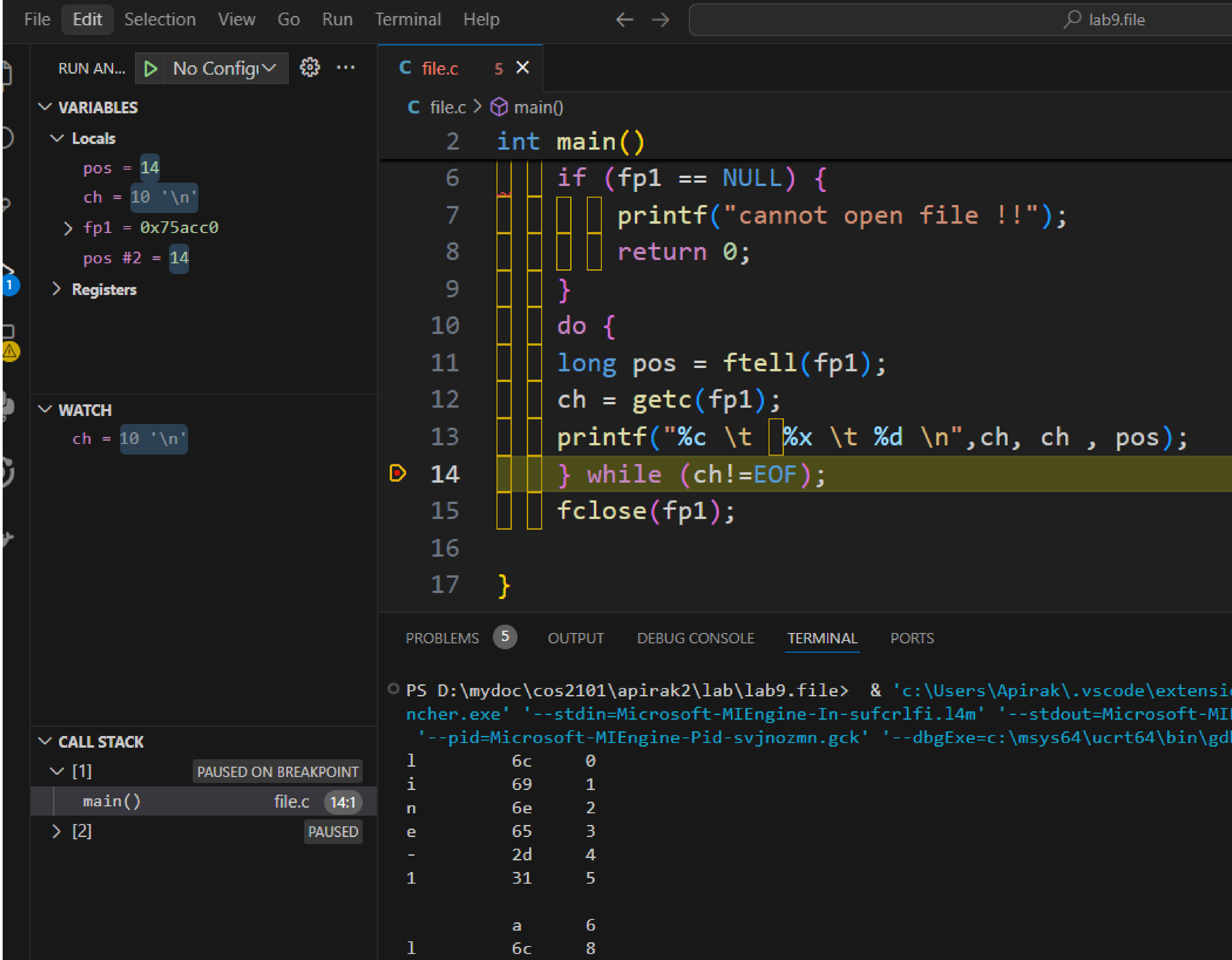
```
00000000: 6C 69 6E 65 2D 31 0D 0A|6C 69 6E 65 2D 32 0D 0A  | line-1..line-2..
00000010: 6C 69 6E 65 2D 33 0D 0A|6C 69 6E 65 2D 34 0D 0A  | line-3..line-4..
```

binary

	a	23
l	6c	24
i	69	25
n	6e	26

text

i	69	25
n	6e	26
e	65	27
-	2d	28
4	34	29



File Edit Selection View Go Run Terminal Help
lab9.file
RUN AN... No Configu
file.c 5
VARIABLES
Locals
pos = 14
ch = 10 '\n'
fp1 = 0x75acc0
pos #2 = 14
Registers
WATCH
ch = 10 '\n'
CALL STACK
[1] PAUSED ON BREAKPOINT
main() file.c 14:1
[2] PAUSED
file.c > main()
2 int main()
6 if (fp1 == NULL) {
7 printf("cannot open file !!");
8 return 0;
9 }
10 do {
11 long pos = ftell(fp1);
12 ch = getc(fp1);
13 printf("%c \t %x \t %d \n",ch, ch , pos);
14 } while (ch!=EOF);
15 fclose(fp1);
16
17 }
PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS D:\mydoc\cos2101\apirak2\lab\lab9.file> & 'c:\Users\Apirak\.vscode\extensio
ncher.exe' '--stdin=Microsoft-MIEngine-In-sufcrlfi.14m' '--stdout=Microsoft-MIE
'--pid=Microsoft-MIEngine-Pid-svjnozmn.gck' '--dbgExe=c:\msys64\ucrt64\bin\gdb
l 6c 0
i 69 1
n 6e 2
e 65 3
- 2d 4
1 31 5
a 6
l 6c 8

ฟังก์ชัน fwrite

int fwrite (void **buffer*, **int** *size*, **int** *count*, **FILE** **stream* **);**

input

buffer Storage location for data

size Item size in bytes (sizeof data type)

count Maximum number of items to be written

stream Pointer to **FILE** structure

return

The **fwrite** function writes up to *count* items, of *size* length each, from *buffer* to the output *stream.*

```c
struct mystruct {
	int i;
	char ch;
	char str[3];
};

int main(void)
{
	FILE *stream;
	struct mystruct s[10] = { 0 };
	if ((stream = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/TEST.$$$", "wb")) == NULL) {
		fprintf(stderr, "Cannot open output file.\n");
		return 1;
	}
	for (int i = 0; i < 10; i++) {
		s[i].i = i;
		s[i].ch = 'A' + i;
		strcpy(s[i].str, "12");
		printf("\n%d, %c , %s", s[i].i, s[i].ch, s->str);
	}
	for (int i = 0; i < 5; i++) {
		fwrite(&s[ i ], sizeof(struct mystruct), 1, stream);
	}

	/* write struct s to file */
	fwrite(&s[5], sizeof(struct mystruct), 5 , stream);
	fclose(stream); /* close file */
	return 0;

}
```

```c
struct mystruct {
    int i;
    char ch;
    char str[3];
};
```

File Edit Options Encoding Help

```
00000000: 00 00 00 00 41 31 32 00|01 00 00 00 42 31 32 00  |....A12.....B12.
00000010: 02 00 00 00 43 31 32 00|03 00 00 00 44 31 32 00  | ...C12.....D12.
00000020: 04 00 00 00 45 31 32 00|05 00 00 00 46 31 32 00  |....E12.....F12.
00000030: 06 00 00 00 47 31 32 00|07 00 00 00 48 31 32 00  |....G12.....H12.
00000040: 08 00 00 00 49 31 32 00|09 00 00 00 4A 31 32 00  |....I12.....J12.
```

ฟังก์ชัน fread

int fread (void **buffer*, **int** *size*, **int** *count*, **FILE** **stream* **);**

input

buffer Storage location for data

size Item size in bytes (sizeof data type)

count Maximum number of items to be read

stream Pointer to **FILE** structure

return

The **fread** function reads up to *count* items of *size* bytes from the input *stream* and stores them in *buffer.*

fread( ) and fwrite( )

size_t fread(void *buffer, size_t numbytes, size_t count, FILE *a_file);

size_t fwrite(void *buffer, size_t numbytes, size_t count, FILE *a_file);

- Buffer in fread is a pointer to a region of memory that will receive the data from the file. Buffer in fwrite() is a pointer to the information that will be written to the file.
- The second argument is the size of the element; it is in bytes. For example, if you have an array of characters, you would want to read it in one byte chunks, so numbytes is one. You can use the sizeof operator to get the size of the various datatypes; for example, if you have a variable, int x; you can get the size of x with sizeof(x);

Contd..

- The third argument is simply how many elements you want to read or write; for example, if you pass a 100 element array
- The final argument is simply the file pointer
- Size_t is an unsigned integer.
- fread() returns number of items read and fwrite() returns number of items written
- To check to ensure the end of file was reached, use the feof function, which accepts a FILE pointer and returns true if the end of the file has been reached.

```
struct mystruct {
	int i;
	char ch;
	char str[3];
};

int main(void)
{
	FILE *stream;
	//read/write where the file pointer location is at begining of file
	if ((stream = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/TEST.$$$", "a+b")) == NULL) {
		fprintf(stderr, "Cannot open output file.\n");
		return 1;
	}

	while (true)
	{
		struct mystruct itm;
		int recordno = 0;

		cout << "\nfinding record  no.  (0=terminate) (1-9) : ";
		cin >> recordno;
		recordno -= 1;
		if (recordno < 0)
			break;

		fseek(stream, recordno * sizeof(struct mystruct), SEEK_SET);
		fread( &itm, sizeof(struct mystruct), 1, stream);

		printf("\n%d, %c , %s -> record no : ", itm.i, itm.ch, itm.str, recordno);

	}
	fclose(stream); /* close file */

	return 0;
}
```

```
#define _CRT_SECURE_NO_WARNINGS
#include <iostream>
#include <stdio.h>
#include <conio.h>

struct prod {
    int cat_num;
    float cost;
};

int main()
{
    FILE *fp;
    struct prod product;

    struct prod a[3] = { {2,20.1},{4,40.1},{6,60.1} };
    struct prod k, *p = &k;

    fp = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/fread1.txt", "w+");
    // write the entire array into the file pointed to by fp
    fwrite(a, sizeof(product), 3, fp);
    // prepare for reading from the beginning of the file
    fseek(fp, SEEK_SET, 0);
    // read from the file one product at a time
    for (int i = 0; i < 3; i++) {
        fread(p, sizeof(product), 1, fp);
        printf(" product %d, cat_num=%d, cost=%f\n", i, p->cat_num, p->cost);
    }
    fclose(fp);
}
```

Rewrite the code saving data file

ตัวอย่าง ให้นักศึกษาเขียนโปรแกรมขายสินค้าและพิมพ์ใบเสร็จรับเงิน
โดยข้อมูลในระบบมีไม่เกิน 100 รายการ
ข้อมูลขายสินค้าด้วย

- รหัสสินค้า(id) เช่น “AB001”
- ชื่อสินค้า(name) เช่น “CAFÉ”
- ราคาขาย(cost) เช่น 30.6
- จำนวนสินค้าในสต๊อก (instock) เช่น 5

Menu

1. Add item to Cart
2. Sell Product
3. Check Stock
4. End Program

Please select 1 / 2 / 3/ 4 =?

กด 1

Input your product data:
id : AB001
name : CAFE
Cost : 30.6
Stock : 5

กด 2

Input your product id: AB001
Input your sell volume : 2
name : CAFE
Cost : 30.6
stock : 3
Pay : 61.2 bath

Menu

1. Add item to Cart
2. Sell Product
3. Check Stock
4. End Program

Please select 1 / 2 / 3/ 4 =?

กด 3

product List:

Id	name	cost	stock
AB001	CAFÉ	30.6	3

กด 4

exit from program

single module Version

```
using namespace std;

struct product {
char id[10];
char name[30];
float cost;
int instock;
};

int main()
{
struct product item[100];
int n = 0;
int select = -1;

while (true)
{

cout << " Menu" << endl;
cout << "1. Add item to Cart " << endl;
cout << "2. Sell product " << endl;
cout << "3. Check stock " << endl;
cout << "4. End program" << endl;
cout << "Please select 1 / 2 / 3/ 4 =? ";
cin >> select;

if (select == 4) {
break;
}
switch (select)
{
case 1:
{
```

แก้ไขให้เป็น dynamic

Exercises

1. Write a C language program to read "mark.dat" file containing rollno, name,marks of three subjects and calculate total mark, result in grade and store same in "result.dat" file. (Note : Make use of fread and fwrite functions)
2. Write a C language program to read a cust.dat file containing meter number, name, current reading & previous reading. Read the same file. Calculate unit and total amounts according to the following rules —

Unit	**rate**
0-50	1.00
51-100	1.50
> 100	2.00

Store meter number ,name ,unit & amount in master.dat file.

END

Lister - [d:\mydoc\cos210

File Edit Options Encodi

```
line-1
line-2
line-3
line-4
```

Lister - [d:\mydoc\cos2101\apirak2\lab\lab9.file\test1.txt]

File Edit Options Encoding Help

```
00000000: 6C 69 6E 65 2D 31 0D 0A|6C 69 6E 65 2D 32 0D 0A  | line-1..line-2..
00000010: 6C 69 6E 65 2D 33 0D 0A|6C 69 6E 65 2D 34 0D 0A  | line-3..line-4..
```

```
File   Edit   Options   Encoding   Help
00000000: 6C 69 6E 65 2D 31 0D 0A|6C 69 6E 65 2D 32 0D 0A  | line-1..line-2..
00000010: 6C 69 6E 65 2D 33 0D 0A|6C 69 6E 65 2D 34 0D 0A  | line-3..line-4..
```

```
l       6c      0
i       69      1
n       6e      2
e       65      3
-       2d      4
1       31      5

        a       6
l       6c      8
i       69      9
n       6e      10
e       65      11
-       2d      12
2       32      13

        a       14
l       6c      16
i       69      17
n       6e      18
e       65      19
-       2d      20
3       33      21

        a       22
l       6c      24
i       69      25
n       6e      26
e       65      27
-       2d      28
4       34      29
```

```c
int main()
{
    char ch;
    FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
    if (fp1 == NULL) {
        printf("cannot open file !!");
        return 0;
    }
    do {
        long pos = ftell(fp1);
        ch = getc(fp1);
        printf("%c \t  %x \t %d \n",ch, ch , pos);
    } while (ch != EOF);

    fclose(fp1);

}
```

```
File   Edit   Options   Encoding   Help
line-1
line-2
line-3
line-4
```

https://www.jdoodle.com/online-compiler-c++14

ฟังก์ชัน ftell

long ftell(FILE *stream*);

input

stream Pointer to **FILE** structure

return

returns the current file position. The value returned by **ftell** may not reflect the physical byte offset for streams opened in text mode

ฟังก์ชัน fseek

int fseek(FILE ****stream*, long** *offset***, int** *origin* **);**

input

stream Pointer to **FILE** structure

offset Number of bytes from *origin*

origin Initial position

- **SEEK_CUR** Current position of file pointer
- **SEEK_END** End of file
- **SEEK_SET** Beginning of file

return

0 if sucessful,

ตัวอย่าง

```
FILE *fp ;

  fp = fseek( fp , 0 , SEEK_SET );    /* go to top of file */
  fp = fseek( fp , 5 , SEEK_SET );    /* go to 5 byte from top file */

  fp = fseek( fp , 0 , SEEK_END );    /* go to button of file */
  fp = fseek( fp , - 5 , SEEK_END );    /* go back 5 byte from button  */

  fp = fseek( fp , 5 , SEEK_CUR  );    /* skip next 5 byte from cur pos */

  fp = fseek( fp , -5 , SEEK_CUR  );    /* go back 5 byte from curs pos*/
```

fseek

```c
int main()
{
    char ch;
    FILE *fp1 = fopen("D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
    if (fp1 == NULL) {
        printf("cannot open file !!");
        return 0;
    }

    do {
        long pos1;
        printf("seek position (-1 exit) : ");
        scanf("%d", &pos1);
        if (pos1 < 0) {
            break;
        }
        fseek(fp1, pos1, SEEK_SET);
        long pos2 = ftell(fp1);
        ch = getc(fp1);
        printf("%c \t  %x \t %d \n", ch, ch, pos2);
    } while (true);

    fclose(fp1);

}
```

NOTE: แฟ้มแบบ text mode

ลักษณะของแฟ้มแบบ text โหมด

1. มี CTRL-Z บอกเครื่องหมายจบแฟ้ม
2. เครื่อง carriage return–linefeed (0xa,0xd) ถูกแทนด้วย linefeed (0xd)
3. การใช้คำสั่งค้นหาตำแหน่งข้อมูล (fseek,ftell) อาจคลาดเคลื่อน
4. ถ้าใช้ฟังก์ชัน **fopen(const char ****filename*, **const char ****mode* **)** ค่าของ mode ต้องมี option " t " หรือไม่ option " b "

ตัวอย่าง

```
FILE  *fp ;

fp = fopen( "c:\myfile.txt" , "rt"  );        → Text File

fp = fopen( "c:\myfile.txt" , "rb"  );        → Binary File
```

Basic File Steps to Handle File

- Opening a file
- Reading data from a file
- Writing data to a file
- Closing a file

```c
int main()
{
    char ch;
    FILE *fp1 = fopen( "D:/mydoc/cos2101/DemoCOS2101/Debug/test1.txt", "rt");
    if (fp1 == NULL) {
        printf("cannot open file !!");
        return 0;
    }
    do {
        long pos = ftell(fp1);
        ch = getc(fp1);
        printf("%c \t  %x \t %d \n",ch, ch , pos);
    } while (ch != EOF);

    fclose(fp1);

}
```

Example: Merge two files

```
#include <stdio.h>
int main()
{   FILE *fileA,  /* first input file */
         *fileB,  /* second input file */
         *fileC; /*  output file to be created */
    int num1,  /* number to be read from first file */
        num2;  /* number to be read from second file */
    int f1, f2;

    /* Open files for processing */
    fileA = fopen("class1.txt","r");
    fileB = fopen("class2.txt","r");
    fileC = fopen("class.txt","w");
```

```
/* As long as there are numbers in both files, read and compare
numbersone by one. Write the smaller number to the output file
and read the next number in the file from which the smaller
number is read. */

   f1 = fscanf(fileA, "%d", &num1);
   f2 = fscanf(fileB, "%d", &num2);

   while ((f1!=EOF) && (f2!=EOF)){
      if (num1 < num2){
         fprintf(fileC,"%d\n", num1);
         f1 = fscanf(fileA, "%d", &num1);
      }
      else if (num2 < num1) {
         fprintf(fileC,"%d\n", num2);
         f2 = fscanf(fileB, "%d", &num2);
      }
      else {  /* numbs are equal:read from both files */
         fprintf(fileC,"%d\n", num1);
         f1 = fscanf(fileA, "%d", &num1);
         f2 = fscanf(fileB, "%d", &num2);
      }
   }
```

```
    while (f1!=EOF){/* if reached end of second file, read
             the remaining numbers from first file and write to
             output file */
        fprintf(fileC,"%d\n", num1);
        f1 = fscanf(fileA, "%d", &num1);
    }
    while (f2!=EOF){ if reached the end of first file, read
             the remaining numbers from second file and write
             to output file */
        fprintf(fileC,"%d\n", num2);
        f2 = fscanf(fileB, "%d", &num2);
    }

    /* close files */
    fclose(fileA);
    fclose(fileB);
    fclose(fileC);
    return 0;
}   /* end of main */
```